

isthmus-ablation-core

Standalone mode

Standalone input
core-only DSMC-style script
ia-core -in sphere.in

Standalone mass source
constant, expression, file
source q1 constant 0.9

Standalone outputs
history, VTU, VTP, .surf
voxel dump hist ...

standalone
loop

Shared core library

Voxel model
IDs, mass, material, fields
voxel create solid sphere ...

Generate/update surface
ISTHMUS mesh + ownership map
isthmus surface surf1 ...

Surface cache
triangle IDs, areas, voxel fractions
map yes

Map surface data
flux/reaction mass to voxels
isthmus map surf1 ...

Ablate voxel mass
local, column, axial, normal carryover
voxel ablate solid ...

DSMC-linked mode

DSMC input
native variables, loops, jumps
label ablate-loop

Converge DSMC interval
particles, collisions, reactions, MPI
run 10000

Surface tallies
reactions/collisions by surface ID
compute ... react/surf

Install new DSMC surface
remap particles and grid-surface state
surface update surf1 install yes

coupled
DSMC loop